ARQ - Mathematical Foundations for Error Valuation

A compact mathematical note on valuing error events through entropy change, utility, cost, and stability risk.

Author attribution: Kotov Ivan
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$$V = \alpha \cdot \Delta H + \beta \cdot U(\rho_{\text{post}}) - \gamma \cdot C - \delta \cdot R(\rho_{\text{post}})$$

This document packages the supplied manuscript into a clean report layout. It keeps the original conceptual core intact: an error is not judged only as a failure, but as a state transition whose value depends on uncertainty, goal alignment, cost, and downstream stability.

Entropy shift	Utility to goals	Operational cost	Stability risk

Scope

The note covers both classical and quantum representations of state, then proposes a single composite valuation function V for detected error events.

Operational reading

A positive valuation means the event may be preserved as an Experience Artifact; a negative valuation means the system should suppress or damp it.

1. Entropy as State Uncertainty

Let the cognitive state of entity c be represented by a probability distribution (classical case) or a density matrix (quantum case) over possible configurations. At any moment, the system carries a baseline entropy H_0 or S_0, reflecting its current uncertainty under memory and real constraints.

Classical definition

$$H(X) = -\sum_x p(x) \log p(x)$$

Shannon entropy measures uncertainty across discrete configurations.

Quantum definition

$$S(\rho) = -\mathrm{Tr}(\rho \log \rho)$$

von Neumann entropy captures uncertainty in the density matrix ρ.

An error event is treated as a perturbation that moves the system from a pre-error state ρ_{pre} to a post-event state ρ_{post}. The key question is not merely whether change occurred, but whether that change increased useful optionality or pushed the system toward incoherence.

2. Value of an Error

The proposed valuation combines four terms: entropy shift, utility of the new state, operational cost, and stability risk. The entropy increment itself is:

Entropy change

$$\Delta H = H(\rho_{\text{post}}) - H(\rho_{\text{pre}})$$

The composite valuation used by the manuscript is shown below.

$$V = \alpha \cdot \Delta H + \beta \cdot U(\rho_{\text{post}}) - \gamma \cdot C - \delta \cdot R(\rho_{\text{post}})$$

α · ΔH — exploration term

Rewards an error when it opens useful new state space instead of merely adding noise.

β · U(ρpost) — goal alignment

Measures how well the post-event state serves the entity's goals or coherence with past experience.

γ · C — cost term

Charges energy, time, privilege usage, and other irreversible expenditures.

δ · R(ρpost) — stability penalty

Penalizes transitions likely to increase incoherence, runaway entropy, or fragile entanglement.

In this notation, α, β, γ, and δ are configurable weights set by the human anchor a. A positive V suggests preservation of the event as an Experience Artifact; a negative V suggests suppression.

3. Measuring Entropy Change in Practice

For classical cognitive flows such as retrieval, ranking, or decision traces, the manuscript suggests approximate measurements rather than insisting on an exact closed form in every case.

Predictive coding

$$H \approx -\sum \log p(\text{observation} \mid \text{current model})$$

If the event reduces prediction error, ΔH may be negative; if it opens new possibilities, ΔH may be positive.

Mutual information

$$\Delta I = I(\text{state; future rewards})$$

Errors that increase mutual information with future rewards are treated as potentially valuable.

Hybrid and quantum reading

For quantum or hybrid systems, S(ρ) may be computed through partial trace over subsystems. An increase in entanglement with the environment can be destructive even when entropy rises, so the risk term may dominate.

4. Decision Rule

Each detected error is classified by computing V and comparing it with promotion and suppression thresholds. The report's logic is deliberately simple so it can be audited and budgeted.

Decision band	Outcome
$V > V_{promote}$	Mint Experience Artifact (EA); preserve the event.
$V < V_{suppress}$	Apply anti-resonance suppression or damping.
Otherwise	Log only; no immediate action.

Thresholds $V_{promote}$ and $V_{suppress}$ are set by the human anchor and can be tightened when energy, time, or privilege budgets are running low.

5. Learning the Valuation Weights

The weights are not assumed to be static. The manuscript proposes a meta-learning loop in which past minted artifacts are revisited and judged against long-term benefit.

1. When an Experience Artifact is minted, its computed value V is recorded together with the witness trail.
2. After a time window, the entity evaluates whether that artifact improved decision quality or reduced later cost.
3. A reinforcement-style update adjusts α, β, γ, and δ so that computed value correlates better with actual long-run utility.
4. Under stable operating conditions, the policy should move toward a better exploration-versus-exploitation balance.

Grounding note

This learning loop stays bounded by physical budgets. In practice that means energy, latency, storage, and privilege escalation are not side details but part of the valuation itself.

6-7. Worked Examples

The original note gives one classical example and one quantum example, showing how the same valuation form produces opposite decisions under different risk structures.

Classical decision error

A decision deviates from the learned policy, but the outcome reveals a useful new path. Here entropy rises moderately, utility is high, cost is small, and risk remains low.

$$V = 0.5 \cdot 0.3 + 1 \cdot 0.8 - 0.5 \cdot 0.2 - 1 \cdot 0.1 = 0.75$$

Result: V = 0.75 -> promote to Experience Artifact.

Quantum decoherence error

A qubit interacts with the environment, entropy rises, utility is near zero, and the risk term dominates because coherence is lost.

$$V = 0.5 \cdot 0.2 + 1 \cdot 0 - 0.5 \cdot 0 - 1 \cdot 0.9 = -0.8$$

Result: V = -0.8 -> suppress.

8. Integration with L4 Budgets

The valuation function is evaluated only when budgets permit. If energy is scarce, γ may be temporarily increased to penalize costly exploration. If the available time window is narrow, a simpler heuristic may be used so that valuation itself does not become the bottleneck.

Earth paragraph

Any real system eventually collides with thermals, latency, memory pressure, and irreversible actions. That is why cost cannot stay abstract: a mathematically elegant policy that ignores power, time, or privilege budgets is not a stable policy - it is just paperwork.

9. Formal Guarantees (Sketch)

Bounded entropy increase	Any error with ΔH above a defined ceiling is always suppressed, preventing divergence into chaos.
Consistency with experience	Minted Experience Artifacts are linked to actual error events through witness trail, reducing the chance of fabricated value.
Convergence	Under stationary conditions, the meta-learning process should move toward a stable exploration/exploitation balance.

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